

Limits at Infinity

Given $\lim_{x \rightarrow \infty} f(x)$ there are four (4) possible values for this limit.

$$\lim_{x \rightarrow \infty} f(x) = \begin{cases} 0 \\ +\infty \\ -\infty \\ \frac{a}{b}, \text{ } a \text{ \& } b \text{ constants} \end{cases}$$

$$\lim_{x \rightarrow \infty} \frac{7x^2 - 4x + 3}{6x^2 - 1}$$

Let $f(x) = \frac{P_m(x)}{Q_n(x)}$, where m and n are the degrees of the polynomials $P(x)$ and $Q(x)$.

Note: To determine the value of $\lim_{x \rightarrow \infty} f(x)$, divide the numerator and the denominator by the highest degree of the denominator.

CASE 1: $m < n$

ex: $\lim_{x \rightarrow \infty} \frac{x+1}{x^2+4}$ $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{4}{x^2}}$$

must show this process!

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{1}{x^2}}{1 + \frac{4}{x^2}} = \frac{0 + 0}{1 + 0} = 0$$

Summary: If $m < n$, $\lim_{x \rightarrow \pm \infty} f(x) = 0$

* H.A. @ $y = 0$

CASE 2: $m > n$

$$\begin{aligned}\text{ex: } \lim_{x \rightarrow \infty} \frac{x^3 + x^2 + 4}{3x^2 + 1} & \frac{\infty}{\infty} \\&= \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^2} + \frac{x^2}{x^2} + \frac{4}{x^2}}{\frac{3x^2}{x^2} + \frac{1}{x^2}} \\&= \lim_{x \rightarrow \infty} \frac{x + 1 + \frac{4}{x^2}}{3 + \frac{1}{x^2}} \\&= \frac{\infty + 1 + 0}{3 + 0} = \infty\end{aligned}$$

$$\begin{aligned}\text{ex: } \lim_{x \rightarrow -\infty} \frac{x^2 + 2}{x - 1} \\&= \lim_{x \rightarrow -\infty} \frac{\frac{x^2}{x} + \frac{2}{x}}{\frac{x}{x} - \frac{1}{x}} \\&= \lim_{x \rightarrow -\infty} \frac{x + \frac{2}{x}}{1 - \frac{1}{x}} \\&= \frac{-\infty + 0}{1 + 0} = -\infty\end{aligned}$$

Summary: If $m > n$ $\lim_{x \rightarrow \pm \infty} f(x) = \pm \infty$

CASE 3: $m = n$

ex: $\lim_{x \rightarrow \infty} \frac{x^3 + 2x + 3}{4x^3 + x^2 + x + 1}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^3} + \frac{2x}{x^3} + \frac{3}{x^3}}{\frac{4x^3}{x^3} + \frac{x^2}{x^3} + \frac{x}{x^3} + \frac{1}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x^2} + \frac{3}{x^3}}{4 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}}$$

$$= \frac{1 + 0 + 0}{4 + 0 + 0 + 0}$$

$$= \frac{1}{4} \quad * \text{ H.A. @ } y = \frac{a}{b},$$

where $\frac{a}{b}$ is the ratio of the leading coefficients.

Summary: If $m = n$ $\lim_{x \rightarrow \pm \infty} f(x) = \frac{a}{b}$